Immunodeficiency 96 (IMD96) — Comprehensive Disease Characterization

MONDO:0030693 · OMIM #619774 · LIG1 (DNA ligase I) deficiency

Prepared as a disease knowledge-base entry. Evidence source types are indicated (human clinical, model organism, in vitro/biochemical, computational/database). Because this is an ultra-rare disorder, the entire primary human literature consists of a small number of reports (~6 patients total); claims are cited to those primary sources plus the founding biochemical and mouse-model studies.

Summary (answer to the research question)

Immunodeficiency 96 (IMD96) is a rare **autosomal recessive inborn error of immunity caused by biallelic hypomorphic/amorphic mutations in *LIG1***, the gene encoding **DNA ligase I**, the principal replicative DNA ligase in dividing mammalian cells. Loss of ligase activity impairs joining of Okazaki fragments on the lagging strand and completion of excision repair, producing replication stress and genome instability that most affect rapidly proliferating lymphoid and erythroid precursors. The clinical picture is a **combined/antibody immunodeficiency of variable severity** — recurrent (mainly viral) respiratory, gastrointestinal and urinary infections from infancy — accompanied by a characteristic laboratory signature of **hypogammaglobulinemia, lymphopenia, increased circulating γδ T cells, and erythrocyte macrocytosis**, with predisposition to growth retardation, photosensitivity and lymphoma.

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1. Disease Information

- **Overview.** IMD96 is a Mendelian, autosomal recessive DNA-repair/DNA- replication defect presenting as an immunodeficiency. It results from partial deficiency of DNA ligase I. Onset of recurrent, usually viral, respiratory infections occurs in infancy/early childhood; gastrointestinal and urinary tract infections also occur. (Source: OMIM/MONDO disease definition; human clinical — Maffucci 2018,

P 30395541 .)

- **Key identifiers.**
- **MONDO:** MONDO:0030693 (immunodeficiency 96)
- **OMIM (phenotype):** #619774
- **OMIM (gene LIG1):** 126391
- **DOID:** DOID:0061066
- **MedGen:** C5676930 · **UMLS:** C5676930
- **Orphanet:** No dedicated ORPHA code is firmly established for "IMD96"; the entity historically overlaps "DNA ligase I deficiency." *(Not confidently available — flag for curator verification.)*
- **ICD-10:** D84.9 (Immunodeficiency, unspecified) / D80.x (predominantly antibody defects) as closest available codes; **ICD-11:** 4A00.x (primary immunodeficiencies). No IMD96-specific code. *(Approximate.)*
- **MeSH:** No specific descriptor; nearest terms "Ligases"/"DNA Ligase ATP" and "Immunologic Deficiency Syndromes."
- **Synonyms / alternative names:** IMD96; "immunodeficiency, autosomal recessive due to LIG1 deficiency"; DNA ligase I deficiency; DNA ligase 1 deficiency; the index cell line/patient is historically referred to as **46BR**.
- **Data provenance:** Aggregated disease-level knowledge derived from a small number of **individual patient case reports** (n ≈ 6) plus biochemical and mouse-model studies — not from large EHR/registry cohorts.

2. Etiology

- **Primary cause — genetic:** Biallelic (homozygous or compound heterozygous) pathogenic variants in *LIG1*. Alleles are amorphic (null-like) or hypomorphic with residual activity; genotype severity tracks clinical/immunologic severity (human clinical/in vitro — Maffucci 2018,

  P 30395541 ;

  Barnes 1992,

  P 1581963).

- **Genetic risk factors:** The disease is monogenic and fully genetically determined; the only "risk factor" is inheritance of two defective *LIG1* alleles. **Consanguinity** and being from a **kindred segregating LIG1 variants** raise recurrence risk (AR inheritance). No established common-variant susceptibility loci or modifier genes are reported.
- **Environmental risk factors:** No environmental cause. However, because cells are hypersensitive to DNA-damaging agents, **exposure to genotoxins/UV/ionizing radiation/alkylating chemotherapeutics** may aggravate cellular pathology (in vitro — Barnes 1992, P 1581963). Photosensitivity is clinically observed.
- **Protective factors:** Retention of **residual LIG1 catalytic activity** (hypomorphic rather than null alleles) is protective — it explains survival and milder phenotypes, since complete loss is embryonic-lethal in mouse (model organism — Bentley 2002, P 11896201). No dietary/lifestyle protective factors are established.
- **Gene–environment interactions:** Genotoxic environmental exposures interact with the underlying repair defect (cells show hypersensitivity to a variety of DNA-damaging agents), plausibly increasing mutation load and cancer risk (in vitro — Barnes 1992, P 1581963). Direct GxE quantification is unavailable.

3. Phenotypes

Frequencies below are the exact n/N from the official HPO annotation of OMIM:619774 (sources: P 30395541 [Maffucci cohort] and P 1581963 [46BR index patient]). Because the total described cohort is ~6 patients, "n/N" is the most precise frequency obtainable.

HPO annotation table (curated, with frequencies):

HPO term	ID	Frequency	Source
Recurrent infections	HP:0002719	5/5	30395541
Decreased circulating IgG	HP:0004315	6/6	30395541 1581963
Decreased circulating IgA	HP:0002720	6/6	30395541 1581963
Decreased circulating IgM	HP:0002850	5/5	30395541
Increased mean corpuscular volume (macrocytosis)	HP:0005518	5/5	30395541
Increased γδ T-cell proportion	HP:0500270	4/4	30395541
Childhood onset	HP:0011463	3/5	30395541
Infantile onset	HP:0003593	2/5	30395541
Multicystic kidney dysplasia	HP:0000003	2/5	30395541
Eczematoid dermatitis	HP:0000964	1/5	30395541
Recurrent lower respiratory tract infections	HP:0002783	1/1	1581963
Recurrent otitis media	HP:0000403	1/1	

P 1581963

| | Growth delay | HP:0001510 | 1/1 |

P 1581963

| | Motor delay | HP:0001270 | 1/1 |

P 1581963

| | Conjunctival telangiectasia | HP:0000524 | 1/1 |

P 1581963

| | Abnormal T-cell proliferation | HP:0031379 | 1/1 |

P 1581963

| | Intellectual disability (ABSENT) | HP:0001249 | 0/5 |

P 30395541

| | Autosomal recessive inheritance | HP:0000007 | — |

P 1581963

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Narrative detail follows.

Infectious / immunologic (clinical signs & laboratory abnormalities) - **Recurrent respiratory infections, usually viral** — infancy/early-childhood onset; core presenting feature. HPO: *Recurrent respiratory infections* (HP:0002205); *Recurrent viral infections* (HP:0004429). - **Gastrointestinal infections / diarrhea** — HP: *Chronic diarrhea* (HP:0002028); *Recurrent gastrointestinal infections* (HP:0004798). - **Urinary tract infections** — HP:0000010. - **Hypogammaglobulinemia** (laboratory) — reduced immunoglobulins across isotypes: **decreased IgG (HP:0004315, 6/6)**, **decreased IgA (HP:0002720, 6/6)**, **decreased IgM (HP:0002850, 5/5)**; near-universal antibody deficiency (Maffucci 2018). - **Lymphopenia** (laboratory) — HP:0001888. - **Increased circulating γδ T cells** (laboratory; distinctive) — HP: *Abnormal proportion of gamma-delta T cells/Abnormal T cell subset distribution* (HP:0011848 / HP:0011840). - **Combined immunodeficiency** in severe cases — HP:0005387 (severe combined immunodeficiency spectrum); severe end required HSCT.

Hematologic - **Erythrocyte macrocytosis** (laboratory; distinctive) — HP: *Macrocytic anemia/ Increased mean corpuscular volume* (HP:0001972 / HP:0005518).

Other organ involvement / dermatologic / renal - **Multicystic kidney dysplasia** — HP:0000003 (2/5; Maffucci 2018) — notable extra-immune feature. - **Eczematoid dermatitis** — HP:0000964 (1/5; Maffucci 2018). - **Conjunctival telangiectasia** — HP:0000524 (1/1; 46BR) — mimics ataxia-telangiectasia (differential-diagnosis clue). - **Recurrent otitis media** — HP:0000403 (1/1; 46BR).

Growth / neurologic / neoplastic (from index patient, 46BR) - **Growth retardation / growth delay** — HP:0001510 (1/1; Webster/Barnes 1992). - **Motor delay** — HP:0001270 (1/1; 46BR). **Intellectual disability is ABSENT** (HP:0001249, 0/5) — helps distinguish from ataxia-telangiectasia. - **Cutaneous photosensitivity / sun sensitivity** — HP:0000992 (Webster 1992). - **Predisposition to malignancy — lymphoma** — HP:0002665 (*Lymphoma*); the index patient died at 19 with lymphoma (Webster 1992, P 1351188).

Phenotype characteristics. Age of onset: **infancy/early childhood** (some features, e.g., growth retardation, congenital/early). Severity: **variable** (mild isolated antibody deficiency → combined immunodeficiency). Progression: **chronic**, with risk of progressive immune compromise and late malignancy. Frequency among affected individuals: infections, hypogammaglobulinemia, lymphopenia, γδ-T-cell increase and macrocytosis were seen in most reported patients (small n).

Quality-of-life impact. Recurrent infections and, in severe cases, need for immunoglobulin replacement or HSCT substantially affect daily functioning; malignancy risk and growth impairment add long-term burden. No formal EQ-5D/SF-36 data exist for this ultra-rare disorder.

4. Genetic / Molecular Information

- **Causal gene: *LIG1*** — DNA ligase 1. HGNC:6598; NCBI Gene 3978; Ensembl ENSG00000105486; OMIM gene 126391; UniProt **P18858**. Locus **19q13.33** (GRCh38 chr19:48,115,444–48,170,654, minus strand). (Computational/database — MyGene/Ensembl; MONDO xrefs.)
- **Pathogenic variants.**
- *Variant class/type:* Predominantly **missense** hypomorphic alleles in the conserved catalytic domain, plus **frameshift/null** alleles. The index 46BR patient was **compound heterozygous** for **p.Glu566Lys (c.1696G>A)** and **p.Arg771Trp (c.2311C>T)** (NM_000234.3); p.Arg771 lies in the adenylation/AMP-binding active-site pocket, and p.Glu566Lys behaves as a near-null

(Barnes 1992,

P 1581963 ;

Webster 1992,

P 1351188).

Maffucci 2018 expanded the spectrum across 3 kindreds. Additional ClinVar Pathogenic/Likely-pathogenic loss-of-function alleles include **c.1244del (p.Thr415fs)** and **c.2444del (p.Leu815fs)**. (Database — ClinVar.)

- *Classification (ACMG/AMP):* Reported disease alleles are **pathogenic / likely pathogenic** (e.g., p.Glu566Lys = Pathogenic in ClinVar), supported by segregation, functional (enzymatic) assays, and cellular repair-deficiency phenotypes.
- *Functional consequence:* **Loss of function** — amorphic or hypomorphic alleles with **variably decreased ligase enzymatic activity** and a **strongly reduced ability to form the enzyme–adenylate intermediate**, leading to **premature release of unligated adenylated DNA** (in vitro — Barnes 1992,

P 1581963 ;

Maffucci 2018,

P 30395541).

- *Allele frequency:* Individual pathogenic alleles are ultra-rare in gnomAD; biallelic loss is extremely rare (carrier frequency not formally established). gnomAD v2.1.1 gene constraint shows LIG1 is **not haploinsufficient** (pLI≈0.004; oe_lof≈0.29, 90% CI 0.19–0.45; missense Z≈0.82), i.e. heterozygous loss is tolerated — consistent with recessive inheritance and healthy obligate carriers (index patient's mother and two brothers).
- *Somatic vs germline:* **Germline**, inherited.
- **Modifier genes:** None established; residual *LIG1* activity itself is the main modifier of severity. Functional redundancy from DNA ligase III (LIG3)/XRCC1 in some repair contexts may partially compensate (biological rationale).
- **Epigenetic information:** No disease-specific epigenetic signatures reported.
- **Chromosomal abnormalities:** None; IMD96 is a single-gene point-mutation disorder (chromosome 19). No aneuploidy/translocation etiology.

5. Environmental Information

- **Environmental factors:** Not causal. Cellular **hypersensitivity to DNA-damaging agents** (UV, ionizing radiation, alkylating agents) means such exposures are biologically relevant aggravators (in vitro — Barnes 1992, P 1581963); photosensitivity is clinically evident.
- **Lifestyle factors:** No established lifestyle contributors; sun protection is prudent given photosensitivity.
- **Infectious agents:** Infections are a **consequence** of immunodeficiency, not a cause. Reported/expected pathogens: **respiratory viruses** predominate; bacterial respiratory, gastrointestinal and urinary infections also occur.

6. Mechanism / Pathophysiology

Ordered causal chain (initiating lesion → clinical manifestation):

1. **Biallelic hypomorphic/amorphic *LIG1* mutations** *lead to* reduced or absent functional DNA ligase I protein / enzymatic activity (in vitro — Maffucci 2018; Barnes 1992).
2. Impaired formation of the **enzyme–adenylate intermediate** *results in* inefficient nick sealing and **premature release of unligated, adenylated DNA** (in vitro — Barnes 1992, P 1581963 ; Maffucci 2018, P 30395541).
3. This *leads to* **retarded joining of Okazaki fragments** during lagging-strand DNA replication and **incomplete excision repair** (in vitro — Barnes 1992).
4. Which *results in* **accumulation of DNA replication intermediates, unligated nicks, replication stress and genome instability** (model organism — Bentley 2002, P 11896201 ; *inferred* to operate similarly in human hematopoietic precursors).

5. Genome instability + replication stress *lead to* **reduced proliferation and survival of rapidly dividing cells**, disproportionately affecting **lymphoid and erythroid precursors** (model organism — Bentley 2002; *inferred* for human lineage-specific effects).
6. Branch A (lymphoid): *results in* **lymphopenia, impaired B-cell/antibody output (hypogammaglobulinemia), skewed T-cell subsets with increased γδ T cells → recurrent viral/bacterial infections and combined immunodeficiency**.
7. Branch B (erythroid): *results in* **impaired erythropoiesis with erythrocyte macrocytosis** (phenocopying impaired DNA synthesis).
8. Branch C (genome-wide, long-term): cumulative genome instability *leads to* **cancer predisposition (lymphoma)**, and in skin, UV-hypersensitivity *contributes to* **photosensitivity**; systemic replication impairment *contributes to* **growth retardation** (human clinical — Webster 1992, P 1351188).

Category detail supporting these steps - Molecular pathways / processes: DNA replication (lagging-strand Okazaki fragment maturation), **long-patch base excision repair**, nucleotide/excision repair completion. GO: *DNA ligation* (GO:0006266), *DNA replication* (GO:0006260), *base-excision repair* (GO:0006284), *DNA repair* (GO:0006281), *lagging strand elongation* (GO:0006273). - **Cellular processes:** Replication stress, cell-cycle impairment, reduced proliferation/survival of precursors, genome instability. GO: *cellular response to DNA damage stimulus* (GO:0006974). - **Protein dysfunction:** Loss/hypomorphic **DNA ligase I** (UniProt **P18858**, 919 aa; ATP-dependent ligase; PCNA-interacting replicative ligase). Domain map: N-terminal disordered regulatory region (1–270) carrying the **PCNA-interacting/ replication-factory-targeting sequence** and CDK phosphosites; central **adenylation (catalytic) domain** with **active-site Lys568** (forms the N6-AMP-lysine enzyme–adenylate intermediate) and AMP/ATP-binding residues at 566, 573, 621, 720, 725, 744; C-terminal **OB-fold DNA-binding domain**. Disease variants map directly onto catalysis: **p.Glu566Lys alters an AMP-binding residue adjacent to catalytic Lys568** (abolishing adenylation — matching the measured loss of enzyme–adenylate formation), and **p.Arg771Trp** disrupts the OB-fold DNA/nick-binding surface. GO cellular component: *nucleus* (GO:0005634), *replication fork* (GO:0005657). PDB: **1X9N** (human LIG1–DNA complex). (Database — UniProt/PDB; in vitro — Barnes 1992.) - **Immune system involvement: Immunodeficiency** (combined + humoral) from impaired lymphocyte development/proliferation and antibody production. - **Tissue-damage mechanism:** Genotoxic stress/genome instability rather than inflammation or ischemia. - **Cell types (CL):** hematopoietic stem/progenitor cell (CL:0000037), T cell (CL:0000084) incl. γδ T cell

(CL:0000798), B cell (CL:0000236), erythroid progenitor (CL:0000038). - **Molecular profiling:** No large omics datasets; functional biochemistry (ligase/adenylation assays) and cellular DNA-damage survival assays are the principal readouts (Barnes 1992; Maffucci 2018).

7. Anatomical Structures Affected

- **Organ/system level: Immune (hematolymphoid) system** is primary — bone marrow, thymus, lymphoid tissues (UBERON:0002405 immune system; UBERON:0002371 bone marrow; UBERON:0002370 thymus). Secondary involvement: **respiratory tract** (recurrent infection; UBERON:0001004), **gastrointestinal tract** (UBERON:0001555), **urinary tract** (UBERON:0011143), **kidney** (UBERON:0002113; multicystic kidney dysplasia in 2/5 — developmental/structural involvement), **skin** (UBERON:0002097; eczema, photosensitivity, conjunctival telangiectasia), and systemic growth.
- **Tissue/cell level:** Hematopoietic tissue; **lymphocytes** (T incl. γδ, B), **erythroid lineage**; skin epithelium (UV sensitivity). CL terms as in §6.
- **Subcellular level: Nucleus** (GO:0005634) — site of DNA replication/repair where DNA ligase I acts (GO cellular component: *replication fork*, GO:0005657; *nuclear replication fork*, GO:0043596).
- **Localization / lateralization:** Systemic (not lateralized); infections and manifestations are bilateral/diffuse.

8. Temporal Development

- **Onset: Infancy/early childhood** for infections; growth retardation and photosensitivity may be evident early. Pattern: **chronic/insidious** with recurrent acute infective episodes.
- **Progression:** Variable — from **stable mild antibody deficiency** to **progressive combined immunodeficiency**. Genome instability confers **late-onset malignancy risk** (lymphoma in the index patient at age 19).
- **Course & duration: Chronic, lifelong.** Recurrent-infection pattern; severe cases progress to transplant dependence.
- **Remission / critical windows:** No spontaneous remission; **HSCT can be curative** for the immune defect. Early diagnosis before severe infections or malignancy is the key intervention window.

9. Inheritance and Population

- **Inheritance: Autosomal recessive** (homozygous or compound heterozygous *LIG1*); parents are typically asymptomatic carriers (Webster 1992: one mutation inherited from the mother and also present in two healthy brothers — carriers).
- **Penetrance / expressivity:** Presumed high penetrance for biallelic damaging genotypes but **markedly variable expressivity/severity**, correlating with residual ligase activity (Maffucci 2018, P 30395541).
- **Genetic anticipation:** Not applicable (not a repeat-expansion disorder).
- **Germline mosaicism / founder effects:** None reported.
- **Consanguinity:** Relevant, as for AR disorders (homozygous cases).
- **Carrier frequency:** Not established; individual alleles ultra-rare in gnomAD.
- **Epidemiology: Ultra-rare** — only ~6 molecularly confirmed patients described (1 index case + 5 in Maffucci 2018). Prevalence/incidence per 100,000 are not calculable and are effectively unknown/<1 in 10^6.
- **Demographics:** Reported across more than one kindred/ethnicity; no strong sex predilection established (index case female). No defined geographic focus.

10. Diagnostics

- **Laboratory tests (LOINC-type):** CBC showing **erythrocyte macrocytosis (elevated MCV)** and **lymphopenia**; **serum immunoglobulins** showing **hypogammaglobulinemia**; specific antibody responses (impaired).
- **Immunophenotyping (flow cytometry): Increased proportion of circulating γδ T cells**, altered T-cell subsets; a distinctive combination with macrocytosis and hypogammaglobulinemia should prompt LIG1 testing (Maffucci 2018).
- **Functional/cellular assays (in vitro): Cellular hypersensitivity to DNA-damaging agents**; **retarded Okazaki-fragment joining**; **reduced DNA ligase I adenylation/ enzymatic activity** — historically used to characterize 46BR (Barnes 1992, P 1581963).

- **Biomarkers:** The **triad (macrocytosis + γδ-T-cell increase + hypogammaglobulinemia)** is a useful clinical flag; premature release of unligated adenylated DNA is a research biochemical marker.
- **Genetic testing (primary confirmatory): WES/WGS or IEI gene panels** including *LIG1*; confirm biallelic variants by **Sanger**. Single-gene *LIG1* sequencing where phenotype is suggestive. CMA/karyotype/FISH generally uninformative (point-mutation disorder).
- **Clinical criteria / differential diagnosis:** Distinguish from **Bloom syndrome** (BLM; the index patient resembled Bloom's but lacked BLM mutation — Webster 1992), other **DNA-repair/genome-instability syndromes** (Fanconi anemia, ataxia-telangiectasia, Nijmegen breakage syndrome), **common variable immunodeficiency**, and combined immunodeficiencies. Macrocytosis + γδ-T-cell increase help separate IMD96 from typical CVID.
- **Screening:** Cascade/carrier testing within affected families; no population newborn screening.

11. Outcome / Prognosis

- **Severity spectrum:** From **mild antibody deficiency** (favorable with Ig replacement) to **severe combined immunodeficiency requiring HSCT** (Maffucci 2018).
- **Mortality/survival:** No formal survival statistics (ultra-rare). Severe, untreated disease carries risk of fatal infection; **the index patient died at age 19 of lymphoma** (Webster 1992), illustrating malignancy-related mortality.
- **Morbidity:** Recurrent infections, growth retardation, treatment burden (immunoglobulin therapy, transplant), and long-term cancer risk.
- **Prognostic factors: Residual LIG1 enzymatic activity / genotype** is the key determinant; earlier diagnosis and definitive therapy (HSCT) improve immune outcomes. No validated QoL instruments for this disease.

12. Treatment

No disease-specific approved drug exists; management follows inborn-errors-of- immunity principles. - **Supportive / pharmacotherapy: Immunoglobulin replacement therapy (IVIG/ SCIG)** for hypogammaglobulinemia; **antimicrobial prophylaxis and aggressive treatment of infections**. NCIT: *Intravenous Immunoglobulin Therapy* (approx.), *Antibiotic Therapy*. -

Definitive/advanced therapy: Allogeneic hematopoietic stem cell transplantation (HSCT) for the severe combined-immunodeficiency end of the spectrum (used in Maffucci 2018 cohort). NCIT: *Hematopoietic Stem Cell Transplantation* (C15431). Gene therapy is conceptually plausible but **not** reported/established for IMD96. - **Caution — genotoxic agents:** Given DNA-repair deficiency and cellular hypersensitivity, use **radiation and alkylating/DNA-damaging chemotherapeutics with caution** (e.g., in transplant conditioning or any cancer therapy) (in vitro rationale — Barnes 1992). - **Pharmacogenomics / personalized:** Genotype (residual ligase activity) informs whether Ig replacement suffices vs. need for HSCT. - **Treatment outcomes / adverse events:** HSCT can restore immune function; standard transplant risks apply, potentially heightened by conditioning-related genotoxic sensitivity. Formal response-rate data are lacking (small n).

13. Prevention

- **Primary prevention:** Not preventable (germline). **Genetic counseling** for AR recurrence risk (25% per pregnancy for carrier couples); **carrier/cascade testing** in affected families; **prenatal or preimplantation genetic testing** where a familial variant is known.
- **Secondary prevention:** Early recognition of the lab triad → early molecular diagnosis → early institution of Ig replacement/prophylaxis or HSCT before irreversible complications; **malignancy surveillance** given lymphoma risk.
- **Tertiary prevention:** Infection prophylaxis, immunizations as appropriate for immune status, **sun protection** (photosensitivity), avoidance of unnecessary genotoxic exposures.
- **Public health / behavioral:** No population-level measures beyond counseling.

14. Other Species / Natural Disease

- **Taxonomy / orthologs:** *LIG1* is highly conserved. Mouse *Lig1* (NCBI Gene 16881; NCBI Taxon 10090); orthologs across vertebrates and lower eukaryotes (CDC9 in *S. cerevisiae*). Evolutionary conservation of the replicative-ligase function is strong.
- **Natural disease in animals:** No well-characterized spontaneous *LIG1* immunodeficiency reported in companion animals/wildlife (OMIA — none prominent).
- **Comparative biology:** Complete loss is **embryonic lethal in mouse** (Bentley 2002, P 11896201),

underscoring conserved essentiality; human patients survive due to residual activity of hypomorphic alleles — a key cross-species contrast.

- **Transmission / zoonosis:** Not applicable (non-infectious genetic disease).

15. Model Organisms

- **Mouse (*Mus musculus, NCBI Taxon 10090):** Lig1 *knockout — two independent null alleles; embryos develop normally to mid-gestation then die from a specific hematopoietic (fetal-liver) defect that is a quantitative proliferation deficiency rather than a lineage block;* Lig1-*null fibroblasts accumulate replication intermediates and show increased genome instability** despite grossly normal repair activity (Bentley 2002, P 11896201). MGI-type resources apply.
- **Cellular models (in vitro):** The human **46BR fibroblast strain** (index patient) and **engineered LIG1-deficient cell lines** demonstrating chemical/ radiation repair defects and reduced ligase activity (Barnes 1992, P 1581963; Maffucci 2018, P 30395541).
- **Phenotype recapitulation:** Mouse null captures the **hematopoietic proliferation defect and genome instability** central to human pathology but is **more severe (lethal)** and does not model the survivable, variable human immunodeficiency (embryonic lethality precludes study of mature adaptive immunity). Human hypomorphic cell lines better model the partial-deficiency disease.

Supported vs. refuted hypotheses

- **Supported:** IMD96 = autosomal recessive **LIG1 (DNA ligase I) deficiency** (MONDO:0030693 / OMIM 619774). Mechanism = impaired Okazaki-fragment ligation + excision-repair completion → genome instability → lymphoid/erythroid proliferation failure. Characteristic labs: hypogammaglobulinemia, lymphopenia, increased γδ T cells, erythrocyte macrocytosis. HSCT curative for severe cases.

- **Supported (residue-level mechanism):** disease variants map onto the LIG1 catalytic pocket — **p.Glu566Lys** hits an AMP-binding residue adjacent to **active-site Lys568** (abolishing the enzyme–adenylate step measured by Barnes 1992), while **p.Arg771Trp** disrupts the OB-fold DNA-binding domain — a direct structural explanation for the loss-of-function biochemistry. gnomAD confirms LIG1 is not haploinsufficient (pLI≈0.004), consistent with recessive inheritance and healthy carriers.
- **Supported (phenotype frequencies):** curated HPO annotations give near-complete penetrance for decreased IgG/IgA (6/6), decreased IgM (5/5), macrocytosis (5/5), increased γδ T cells (4/4) and recurrent infections (5/5); intellectual disability is explicitly absent (0/5), and renal (multicystic kidney dysplasia 2/5) plus dermatologic features broaden the spectrum.
- **Refuted / corrected:** Initial assumption that "Immunodeficiency 96" was the REL/c-Rel disorder was **wrong** — c-Rel deficiency is **Immunodeficiency 92 (IMD92)**. This was corrected by resolving MONDO:0030693 to LIG1.
- **Refuted historically:** The index patient was NOT Bloom syndrome despite clinical resemblance (Webster 1992) — a distinct genetic entity.

Limitations & future directions

- **Ultra-rare** with ~6 molecularly confirmed patients; frequencies, penetrance, survival and QoL are not quantifiable. Orphanet/ICD-specific codes are uncertain and need curator confirmation.
- No omics (transcriptomic/proteomic/metabolomic) disease datasets; no dedicated gene therapy program.
- Future work: define exact recurrent *LIG1* alleles and genotype–phenotype/ residual-activity correlations; systematic malignancy-risk surveillance; conditional/hypomorphic mouse or patient-iPSC models to study lineage-specific immune defects and to test safer (reduced-genotoxicity) transplant conditioning.

Key references (PMID)

- **30395541** — Maffucci et al. 2018, *J Clin Invest*: biallelic LIG1 mutations underlie a spectrum of immune deficiencies (5 patients/3 kindreds). *(Human clinical + in vitro.)*
- **1351188** — Webster et al. 1992, *Lancet*: growth retardation and immunodeficiency with LIG1 mutations (index patient; lymphoma at 19). *(Human clinical.)*
- **1581963** — Barnes et al. 1992, *PNAS*: LIG1 mutations in 46BR; Okazaki- fragment/excision-repair defect; reduced enzyme-adenylate formation. *(In vitro/ biochemical.)*
- **11896201** — Bentley et al. 2002: *Lig1*-null mouse; embryonic-lethal hematopoietic proliferation defect and genome instability. *(Model organism.)*
